

Syllabus

Spring 2026

CSC 436: Software Engineering

Live Online, MW 12:30 PM – 1:45 PM



Course Description

Software engineering is as much about teamwork as it is about technology. This course covers both.

Students will learn how software is developed in industry by adopting agile methods, discovering requirements, designing modular systems, selecting effective tests, and using metrics to track progress. The course will also cover the why's behind the how-to's, to prepare students for advances in engineering practices. We will explore how to elicit what users really want, how a clean architecture divides and conquers the inherent complexity of software systems, how test coverage is essential for detecting the inevitable defects in code, and more. Concepts will be illustrated by real-life examples and case studies.

The course includes a significant team project. Students will form teams of 4, where possible, and choose a project aimed at providing value for a user or users outside the team.

Course Catalog Description: Introduces fundamental software engineering techniques for developing correct, efficient, reliable, maintainable, and evolvable software for large software projects. The class project gives an experience in using evolutionary software development processes for developing a high-quality, evolvable software product that addresses emerging customer requirements. Graduate-level requirements include extra reading assignments on advanced topics and home assignments on these readings.

Instructor and Contact Information

- Instructor: Ravi Sethi
- Email: rsethi@arizona.edu
- Office Hours: By appointment

- TA: Phyllis Spence
- Email: pspence1234@arizona.edu

Course Format and Teaching Methods

The course includes lectures, a team project, and team presentations.

For the project, students will form their own teams of 4, prepare a written team proposal, design and implement a system, reflect on their experience, and complete a comprehensive final written report. Software and documentation produced by students will be made available as is, under the MIT open-source license <https://opensource.org/license/mit/>.

Live Online Lectures

This class is scheduled to be taught in the **LIVE ONLINE** modality.

Meeting Times: Monday and Wednesday, 12:30 PM – 1:45 PM

- Join URL: <https://arizona.zoom.us/j/85285774616>
- Password: 436206

Course Objectives

- Cover principles and practices related to software engineering topics such as requirements, agile development, continuous deployment, software architecture, software quality, and project management.
- Provide students with the experience of participating in a group project.
- Assign readings from the scientific literature for students to explain and present.

Expected Learning Outcomes

Students who successfully complete this course would be able to:

- Explain the fundamental principles and practices of software engineering.
- As part of a group project: propose, design, and build a software system that meets potentially changing customer requirements.
- Explain their own work and evaluate its contributions through oral presentations and written reports.

Transferable Career Skills

National Association of Colleges and Employers (NACE) Career Readiness:

Career readiness is a foundation from which to demonstrate requisite core competencies that broadly prepare the college-educated for success in the workplace and lifelong career management. For new college graduates, career readiness is key to ensuring successful entrance into the workforce.

There are eight career readiness competencies, each of which can be demonstrated in a variety of ways." (NACE, 2025)

- Career & Self Development

- Communication
- Critical Thinking
- Equity & Inclusion
- Leadership
- Professionalism
- Teamwork
- Technology

In this course, we will focus on the following competencies:

- Career & Self Development – The semester-long project prepares students for software development in the workplace. Students create a software product to meet the needs of real customers.
- Teamwork – We will cover the process of forming and developing a high performing team. We will also cover basic principles and best practices for cooperatively planning and reviewing the progress of a software project.
- Technology – The four main topics of this course are software development methods, user and product requirements, software design and architecture, and software quality.

Makeup Policy for Students Who Register Late

Students who wish to register late must contact the instructor during the first week of classes. Students will be able to submit work for the first week of class if they have registered late.

Course Communication

The class D2L site has general information about the course, including the syllabus.

Announcements, events, homework, project templates, notes, readings, and other materials related to the course will be posted during the semester to the D2L site.

All submissions will be through the D2L course site.

Required Texts or Readings

Students will be assigned readings that will be available on-line either through the University of Arizona Library or through the class course materials page.

Edited versions of the slides from the lectures will be posted to the class course materials page on D2L before the next lecture.

All materials made available through the instructor or The University of Arizona remain the property of the copyright holders. They are provided for the use of students in this course for the duration of the course, except as noted.

Required or Special Materials

For this class you will need daily access to the following hardware: a laptop or web-enabled device with webcam and microphone; regular access to reliable internet signal; ability to download and run the following software: a web browser, Zoom, Adobe Acrobat, etc.

For the project, you may need additional software based on the needs of the project chosen by your team. Students are responsible for providing their own software and hardware systems and services that they might need for their project.

Assignments, Reports, Exams, and Other Graded Items: Schedule/Due Dates

All assignments and reports will be due by 5 PM on the assigned date. The number and due dates of homework assignments are subject to change with advance notice, as deemed appropriate by the professor.

Homework Assignments:

01/23 HW1

02/06 HW2

02/27 HW3

04/03 HW4

04/24 HW5

Project:

01/30 Form Teams

02/16 Project Proposal Presentations Day 1

02/18 Project Proposal Presentations Day 2

02/20 Project Proposals Due

03/20 Status Report 1 Due

03/23 Status Report 1 Presentations

03/27 Peer Review 1 Due

04/10 Status Report 2 Due

04/13 Status Report 2 Presentations

04/27 Final Project Presentations Day 1

04/29 Final Project Presentations Day 2

05/01 Final Project Reports Due

05/06 Peer Review 2 Due

Weekly Questions: Weekly questions can be found under the Quiz tab on D2L and are due at 5 PM every Friday unless otherwise specified

Attendance: Attendance is required at every lecture.

Exams:

03/04 Midterm Exam (12:30 PM – 1:45 PM)

05/13 Final Exam (10:30 AM – 12:30 PM)

Final Examination

The Final Exam will be held on Wednesday, May 13th from 10:30 AM - 12:30 PM. Please make every effort to avoid conflicts with these times. For exceptions, please contact the instructor well in advance of the scheduled exam times.

University Final Exam Schedule: <https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams>

Grading Scale and Policies**Grading Policy:**

Assignments and the Midterm Exam will be graded within 2 weeks after the submission date.

Homework grades will be returned before the next homework is due

The Final Exam will be graded within 48 hours of the submission time, as per University Policy.

The contributions toward the grade in this course are as follows:

18% - Homework

20% - Midterm

28% - Final

30% - Project

2% - Attendance

2% - Weekly Questions

Grade Distribution for this Course:

A: 90+%

B: 80-89.99%

C: 70-79.99%

D: 60-69.99%

E: 0-59.99%

University Policy Regarding Grades and Grading Systems:

<https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at <https://catalog.arizona.edu/policy/courses-credit/grading/grading-system>.

Dispute of Grade Policy

The dispute period is one week after the grades are posted, after which there will be no changes to the grade.

Honors Credit

Students wishing to contract this course for Honors Credit should e-mail me to set up an appointment to discuss the terms of the contact and to sign the Honors Course Contract Request Form. The form is available at <http://www.honors.arizona.edu/honors-contracts>

Scheduled Topics

This course is an introduction to the following topics in Software Engineering:

- What is Software Engineering
- Introduction to Software Development Processes
- Working with Customers
- Goals and Metrics
- Project Planning
- Design and Architecture
- Static Analysis and Code Reviews • Testing
- Project Management

The ordering of the topics is subject to change. Additional topics may be added if time permits.

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Also watch the video available at:

https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spf-url=common%2Flearningeventdetail%2Fcrtfy0000000000003841

University-Wide Policies

Links to the following UA policies are provided here: <https://catalog.arizona.edu/syllabus-policies>

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy
- Class Recordings
- Additional Resources
- Preferred Names and Pronouns

Department-wide Syllabus Policies and Resources

Links to the following departmental syllabus policies and resources are provided here: <https://www.cs.arizona.edu/cs-course-syllabus-policies>

Department Code of Conduct

Illnesses and Emergencies

Obtaining Help

Confidentiality of Student Records

Land Acknowledgement Statement

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.